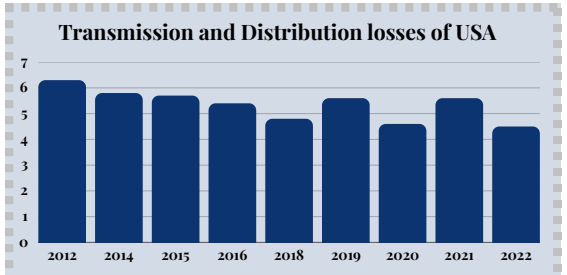
NETAJI SUBHAS UNIVERSITY OF TECHNOLOGY

REDUCING TRANSMISSION LOSSES TO <2% BY 2030



GOALS AND OBJECTIVES



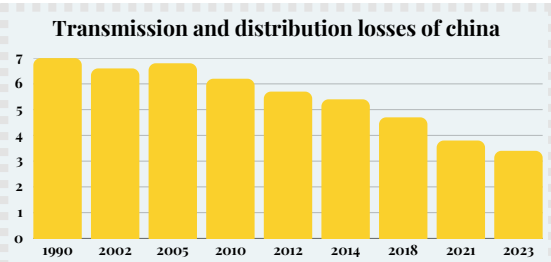


Transmission losses were approximately 2%, demonstrating a very efficient power grid.

2%

Transmission losses in FY 2023-24



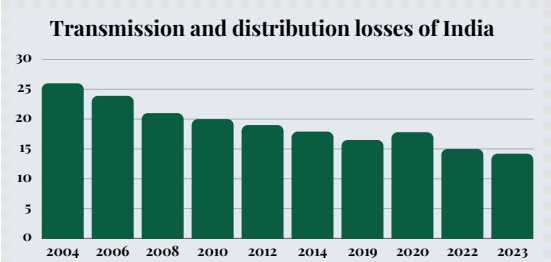


The country maintained low losses at about 1.8%, demonstrating strong transmission infrastructure efficiency

~1.8%

Transmission losses in FY 2023-24





Showed improving but relatively higher grid losses in FY 2023–2024, with transmission losses of roughly 3.55%.

3.55%

Transmission losses in FY 2023-24

OUR GOALS

Benefits for Social and Energy Security: To advance energy equity, cut down on **theft and unmetered consumption**.,two main causes of commercial losses.

Technological and Regulatory Modernization: To ensure fair access in rural areas by **enforcing regulations, and encouraging innovation** to handle variable renewables.

Enhanced Production and Grid Reliability: Goals center on **optimizing generation** by lowering the need for excess production in order to meet rising demand, which is expected to reach 2,500 BU by 2030.

Transmission Loss Reduction Strategies

CHINA

- China's lower losses reflect more efficient infrastructure, tighter controls, better equipment or maintenance (or a combination). The main reason for reducing transmission losses in China is the use of **UHV (Ultra High Voltage)** systems for both AC and DC (800kV DC, 1000kV AC), power sector reforms, renewable integration, and a low-carbon focus.

USA

- Advanced reconductoring doubles or triples capacity sources, and reduces congestion by substituting **high-performance conductors** for lines.
- **FACTS Devices**: Pilots demonstrate real-time improvements by enabling dynamic electricity rerouting for optimal performance and decreased losses.
- **Smart Grid & Infrastructure Replacement**: Replaces aging infrastructure to improve resilience. Uses sensors and automation for monitoring.

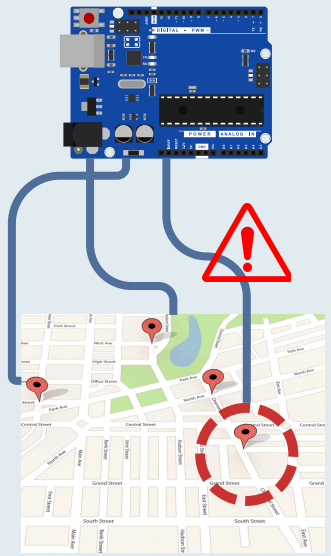
MAIN OBJECTIVES

- Install **AI-driven meters for theft detection** to reduce commercial losses by 50% in pilots with scalable, low-cost rollout under RDSS taking inspiration from China's controls and the USA's smart grids.
- Adopt **Advanced Reconductoring**: Taking a cue from the United States, which has upgraded its existing lines using high-performance conductors to reduce I^2R losses by 20–30%.
- Implement **Market Reforms**: Using China's economic dispatch and interregional trade to improve grid utilization and cut over-generation losses by 10–15% with low upfront costs.
- Establish **HVDC Corridors**: Model China's UHV/HVDC for long-distance transmission, focusing on high-loss areas such as northeast India, and hitting <2% losses with incremental investments that result in yearly fuel savings of Rs 10,000–20,000 crore.

SOLUTIONS AND STRATEGIES

SMART METERING WITH ANTI-THEFT SYSTEM

- **Smart Hardware**: It uses an Arduino, a small computer chip, and sensors to measure how much electricity is flowing to homes.
- **Google Maps Integration**: Every registered customer's meter is shown on an interactive Google Map.
- **The "Red Circle"**: When someone connects an illegal wire to the power line, the system quickly detects a new connection that isn't in its database. It marks that location on the map so the company knows exactly where the theft is happening.



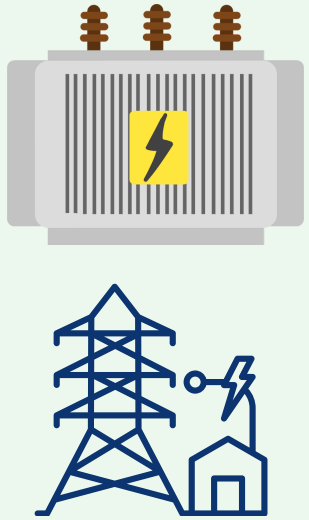
POWER REFORM POLICY

- **Plug-and-Play Standards**: Standardize smart meters to cut down on long-term expenses and prevent vendor lock-in.
- **Time-of-Day Pricing**: To influence customer behavior, provide reduced off-peak rates with shadow bills.
- **Direct Subsidy** (DBT): To guarantee immediate utility payments and enhance grid investment, DBT transfers subsidies straight to prepaid accounts.



TECHNICAL LOSS REDUCTION

- **Smart Power Bridges** (Soft Open Points): The system employs smart switches which maintain optimal load distribution while preventing power outages through automatic overload protection.
- **Digital Transformers** (SSTs): The system employs solid-state transformers instead of traditional transformers which results in greater energy savings.
- **AI Grid Brain** (Digital Twin): The system uses AI-powered grid simulations to forecast demand and manage voltage levels while achieving a 24 percent improvement in operational efficiency.



RECONDUCTORING AND FACTS DEVICES

- **Use ACCC** to convert conductors to reconductoring: To reduce I^2R heat losses by 20–30% without building new transmission structures, swap out old conductors with low-resistance ACCC.
- **Use FACTS** for Dynamic Control: Set up devices such as STATCOM to control voltage, minimize 10–20% overload losses, and enhance the integration of renewable energy sources into the grid.
- **Target High-Loss & Renewable Corridors**: To optimize efficiency gains and return on investment, integrate both technologies in high-demand and high-loss areas, such as Arunachal Pradesh, Nagaland, Gujarat, and Rajasthan.



GRID LOSS MINIMIZATION STRATEGY: MARKET EVALUATION

Sector Recovery & Financial Turnaround

- National AT&C losses reduced from **22.32% FY21** to an unprecedented **15.04% FY25.**
- DISCOMs achieve first-ever collective profit of ₹2,701 Cr in 2025.
- ACS-ARR gap reduces to ₹0.06/kWh, indicating complete cost recovery for utilities.

Legislative & Policy Foundations

- **Electricity Bill 2025** mandates DBT to stabilize utility cash flows and ensure subsidy protection.
- Statutory 5-year period introduced to completely eliminate industrial cross-subsidies.
- 'Non-Fossil' purchase obligation introduced for utilities to drive RE integration.

Scaled Infrastructure Deployment

- RDSS Budget of ₹3.04 Lakh Cr currently active with 45 utilities and 28 states covered until 2026.
- **5.28 Cr smart meters already installed** by late 2025 (towards **25 Cr national target**).
- Authorized infrastructure works now cover 52.5 Lakh DTs and 2.05 Lakh feeders for complete grid visibility.

The Efficiency Revenue Nexus (2% Goal)

- A mere 2% loss reduction potential unlocked ₹18,450 Cr of additional revenue.
- GENCO outstanding dues cleared by 96% with ₹4,927 Cr still pending.
- High-loss states of Bihar and UP now **prioritizing smart prepaid options** to achieve 99%+ collection

Technology & Supply Chain Readiness

- ACCC conductors now doubling capacity with 214 ckt-km urban corridors (case study - Delhi).
- **CEA 2025 Interoperability Guidelines** now mandate vendor-agnostic AMI system architecture.
- Supply chain scaling for high-efficiency hardware with CAGR of 6.0% in HTOC market.

Strategic Intervention Matrix: Market Readiness & Economic Opportunity (2025–2030)

METRIC	STRATEGIC	TECHNICAL	COMMERCIAL
READLINESS	TRL 6-7 SSTs, AI digital Twin adoption in R&D hubs	Proven 175,000 km ACCC installs,strong local BHEL/Sterlite mfg	Sanctioned for 45 utilities, 2025 DBT policy support
LOSS IMPACT	AI optimisation outperforms traditional load flow algorithms	ACCC reduces line losses by 25-40% vs conventional ACSR.	AI-driven detection of theft and billing anomalies.
5-YEAR TAM	Projected CAGRof 13.2% for specialised smart tech to 2030.	Sanctioned RDSS allocation for physical loss reduction works.	Sanctioned RDSS cost for smart metering works.

EXECUTION PLAN

SMART METERING



Install smart meters for energy balancing on feeders and DTs.

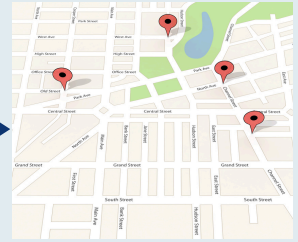
Smart Meter Installation



Set up a secure central Meter Data Management System (MDMS).
Link each smart meter to:

- Consumer ID
- Feeder
- Transformer

Central Database Setup (MDMS)



- Geo-tag all network assets with GPS devices.
- Create digital GIS maps of entire distribution networks.
- Link every consumer ID with GIS location and feeder polygons.
- Enable feeder-wise energy comparison and loss identification.

GIS Mapping & Asset Linking



Monitor smart meter data for constant alerts of possible tampers and anomalies.

Monitoring & System Management

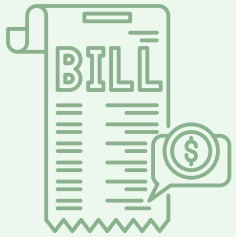
POLICY REFORM

- Define plug-and-play technical standards
- Deploy compatible smart meters



Standardization & Infrastructure Setup

- Introduce shadow billing
- Educate consumers on the benefits of ToD pricing



Consumer Awareness & Pilot Testing

- Switch on ToD pricing
- Integrate subsidy DBT with prepaid meter accounts



Implement ToD Pricing & DBT Integration

- Utilities receive real-time payments
- Improve the grid through reduced technical losses



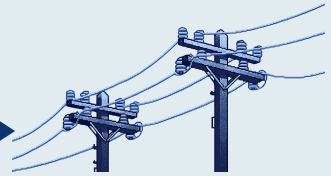
Revenue Stabilization & Grid Improvement

ADVANCED CONDUCTORS



Use smart meter + GIS + feeder data to find heavy loss circuits.

Data-Driven Hotspot Identification



Cost effective reconductoring

Strengthen lines where they are most needed and replace outdated conductors with HTLS.Instead of merely enlarging the conductors, use high-temperature low-sag (HTLS) conductors, which offer a greater current capacity at a lower right-of-way cost.



To control voltage and flows, install STATCOM/SVC/UPFC at vulnerable grid locations.Prioritize high-impact feeders over system-wide installation.

Selective FACTS Deployment

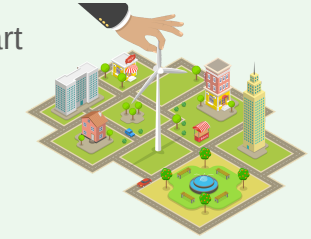


By 2030, integrate analytics and monitoring, and track results

Phased Rollout with Smart Controls

TECHNICAL LOSS REDUCTION

Install smart bridges in the main network points.



Infrastructure upgrade

Replace conventional transformers by SSTs gradually.



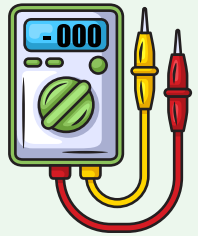
Transformer Modernisation

Digital twin model of the grid.
Include data from the monitoring in real time.



AI Integration

It balances load and voltage using AI.
Continuous technical losses reduction.



Automated Optimization

ROADMAP

	PHASE 1(0-12 MONTHS)		PHASE 2(12-30 MONTHS)		PHASE 3(30-50 MONTHS)	
	IDENTIFICATION	BASIC FRAMEWORK	EXECUTION	TARGETED UPGRADES	OPTIMISATION	ENFORCEMENT
ADVANCED CONDUCTORS	**Identify overloaded transmission lines** and high-loss corridors using load flow studies, positive real-time SCADA data, and where there are numerous renewables.	Concentrate first on the lines where upgrading conductors results in 20-30% reductions in losses **without requiring new tower construction.** Prepare DPRs and secure funding throughH RDSS and PGCIL planning routes.	ACSR to be replaced by HTLS or **ACCC on priority corridors**. Reduce line loss and thermal overload conditions. Improve transfer capacity using existing pole structures.	Construct FACTS devices (i.e. STATCOM or SVC) at **weak nodes** to improve both voltage profiles & reactive power flows.	Increase the number of high-load Corridor reconducting Establish **dynamic load balancing.** Maximize Power Factor Corrections	Reduce System Technical Losses by an Order of Magnitude Work with HVDC for bulk transfers. **Maximize corridor efficiency**
HVDC LINES	Identify long-distance high load corridors between renewable-rich states and industrial and metro demand centers. Congested AC corridors are prioritized for detail Project Reports (DPRs) that utilize load flow/stability analysis	Begin the **construction of HVDC converters** to support the HVDC bipole for reliable operation. Plan for integration and **coordination of existing HVAC** grids and future DC grids so as to avoid instabilities in either.	Coordinate commissioning with **HVAC load shifting**. Create the integration of operational system real-time (HVDC) controller with EMS.	Train operational personnel on HVDC controller operation Begin phased energisation.	Prepare to operate HVDC at full capacity Transition the transfer of bulk renewable energy to HVDC **Decommission the use of HVAC** for Durable/Overloaded lines	Improve the Grid Stability Measures Connect **HVDC to National Dispatch Centers** Minimize the transmission losses associated with long-distance transmission.
SMART METERING	Begin with smart meter installations at the **locations of highest loss** in urban and industrial areas of the RDSS program. Start on the creation of a GIS map of the feeders, transformers, poles, and customers complete with geo-tags.	**Integrate SCADA base with MDMS and GIS**. Create a dashboard of real-time feeder loss monitoring providing monitoring data on an integrated basis for the determination of weak areas.	Create heat loss maps. Install outage management system (OMS). Provide capacity for real-time **voltage and current tracking.**	Improve automation of **fault detection and isolation** capabilities. Begin to establish predictive maintenance analytics.	**Full virtual representation** of the entire electrical network (Digital Twin) Automatic reconfiguration of feeders through the application of technology	Continuous GIS updates from use of field applications Funding must be linked to **measureable reductions in losses**
POLCY REFORMS	Preparation of **Direct Subsidy Scheme** Consumers of Subsidy (Agriculture, BPL & OTHERS) will be identified for the purpose of subsidy under Direct Benefit Transfer scheme.	**Digital mapping** using GIS-based digital mapping and Aadhaar linked to the consumers database.Establishing the process for Direct Benefit Transfer to rectify previously distorted tariff.	Launch **Time of Use (TOU) rates**. Identify tampered connections or unusual load profiles. Expand to prepaid billing systems in metro cities.	Establish **strict billing enforcement in high loss areas**. Achieve >95% collections within 30 days. •Eliminate commercial loss hot spots (high loss locations).	Compulsory use of **prepaid smart meters** and bills in urban areas 98% - 99% collection efficiency Eliminate late billing processes	**Dynamic pricing** (based on renewable sources) for consumers AT&C losses at the bare minimum Transparent tracking of consumption for consumers

BETTER GRID. BETTER ECONOMY. BETTER TOMORROW.

We Fix the Root Cause, Not Just the Symptoms
Instead of blindly building new infrastructure, we start with smart metering + GIS analytics to identify exactly where losses occur ensuring every rupee is invested with precision.

Policy Reform Reduces Technical Stress
Time-of-Use tariffs shift demand away from peak hours, lowering current flow and directly reducing I^2R transmission losses without physical expansion.

Immediate Financial Recovery Before Heavy CAPEX
Anti-theft systems, shadow billing, prepaid metering, and DBT improve collection efficiency quickly, generating internal revenue to support grid upgrades.

Reconductoring Delivers High Impact at Lower Cost
By upgrading existing conductors (HTLS/ACCC), we increase ampacity and cut resistance losses without land acquisition or new tower construction.

FACTS Stabilize and Optimize the Grid
STATCOM/SVC devices dynamically control voltage and reactive power, preventing overload losses and ensuring renewable stability.

HVDC Builds a Future-Ready Backbone
Strategic HVDC corridors enable bulk, low-loss renewable power transfer over long distances reducing congestion and strengthening national grid reliability.

Balanced, Phased, and Economically Sustainable
Low-cost digital reforms fund medium-cost upgrades, while high-investment HVDC is deployed only where justified creating a financially realistic pathway to <2% transmission loss.

Reduced congestion

Stopped theft and non payment

Gaining Public trust

<2%
TRANSMISSION LOSSES

LOSS REDUCTION → **POLICY REFORM** → **CAPACITY UPGRADES** → **BULK TRANSFER** → **2030**

Reduced I^2R losses

Reduced technical and non technical losses

Transmission losses reduced to 2% by 2030

IMPACT, FEASIBILITY AND SCALABILITY

	SOCIAL IMPACT	ECONOMIC IMPACT	ENVIRONMENTAL IMPACT	TECHNOLOGICAL IMPACT
FOCUS	Consumer Equity & Reliability	DISCOM Profitability & Rural Economy	Decarbonization & Resource Efficiency	Grid Intelligence & Transparency
KEY RESULT	• 20% Quicker Restoration • 12% Peak Shifting • Secured DBT Subsidy	• ₹2,701 Crore Profit • 15.04% AT&C Losses • ₹0.06/kWh Revenue Gap	• 130,000 Tonnes CO2 • 40% I^2R Reduction • 30% Emission Cut	• 98.3% Fault Accuracy • 50% SST Shrinkage • Plug-Play AMI Standard

FEASIBILITY

National Funding: ₹3.03 Lakh Crore sanctioned for RDSS, out of which ₹1.53 Lakh Crore is for Infra and ₹1.31 Lakh Crore for AMI.
Technology Maturity: GIS mapping and reconductoring through ACCC have achieved TRL 9, having been deployed in 165+ projects. AI Digital Twning and SSTs have achieved TRL 6-8, with live pilots in Janakpuri and IIT Roorkee.
Operational Evidence: DHBVN, Haryana, has achieved consistency in profitability through a ₹1.2 Billion investment in IT/GIS, mapping 433 transformers and thousands of poles.
Risk Mitigation: A minimum of 60% local content required for smart meter manufacturing to mitigate global supply chain shocks.

SCALABILITY

Rollout Target: National rollout of 250 million smart meters by 2027; currently, we have 5.44 Crore smart meter installations across 28 states in India (Jan 2026).
Industrial Capacity: Domestic production capacity for high-performance conductors is over 150,000 MT, and power cables exceed 3,000 km.
Legislative Backbone: Electricity Amendment Bill 2025 has provided the legislative backbone for shared network competition and cost-reflective tariffs, removing "progressive" transition clauses.
Supply Constraints: Scalability is restricted by the availability of Silicon Carbide (SiC) semiconductors for SSTs and carbon fiber cores for reconductoring through ACCC.

ANNEXURE

WORKING MODEL AND ITS CODE

CODE

```
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>

#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 32
#define OLED_RESET -1

Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, OLED_RESET);

#define LOAD1 2
#define LOAD2 3
#define BUTTON 7

long balance = 5000;
unsigned long previousMillis = 0;
const long interval = 100;

void setup()
{
  pinMode(LOAD1, OUTPUT);
  pinMode(LOAD2, OUTPUT);
  pinMode(BUTTON, INPUT_PULLUP);

  digitalWrite(LOAD1, HIGH);
  digitalWrite(LOAD2, HIGH);

  if(!display.begin(SSD1306_SWITCHCAPVCC, 0x3C))
  {
    while(true);
  }
}

void loop()
{
  if(digitalRead(BUTTON) == LOW)
  {
    balance = 5000;
    digitalWrite(LOAD1, HIGH);
    digitalWrite(LOAD2, HIGH);
    delay(300);
  }

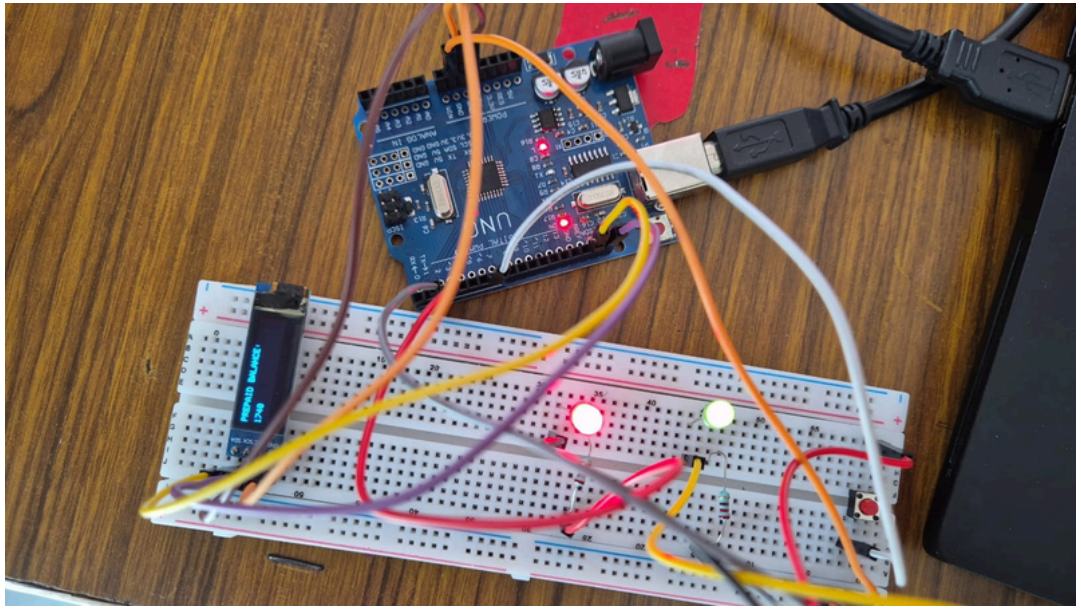
  unsigned long currentMillis = millis();

  if(currentMillis - previousMillis >= interval)
  {
    previousMillis = currentMillis;

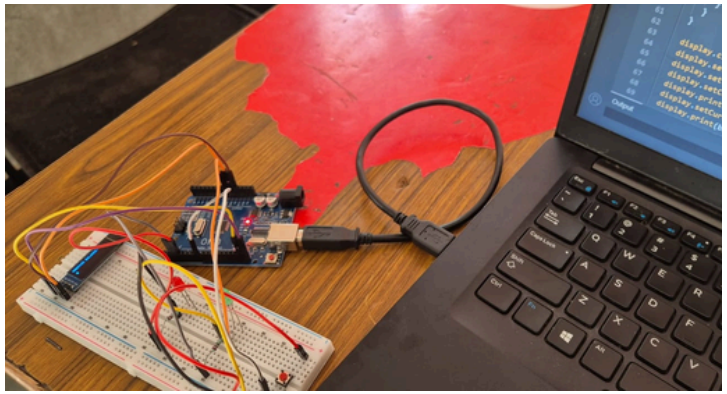
    if(balance > 0)
    {
      balance -= 20;

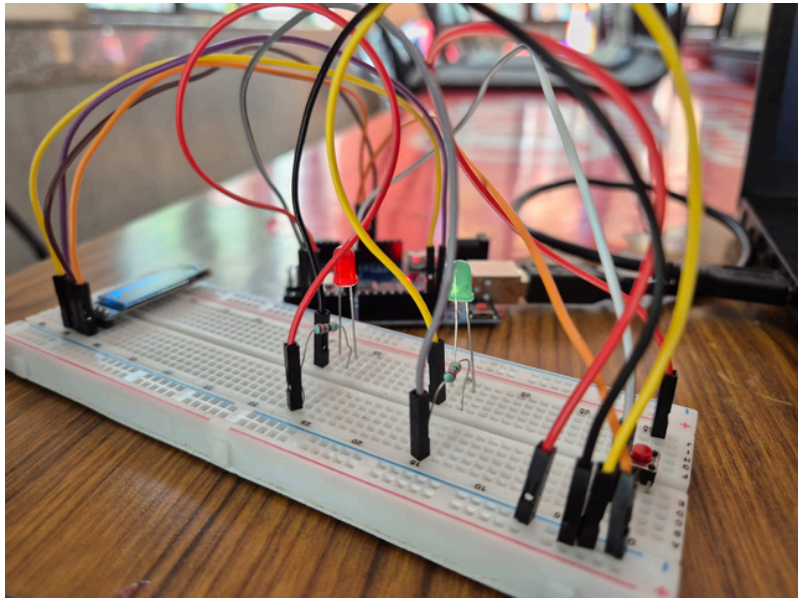
      if(balance <= 0)
      {
        balance = 0;
        digitalWrite(LOAD1, LOW);
        digitalWrite(LOAD2, LOW);
      }
    }
  }

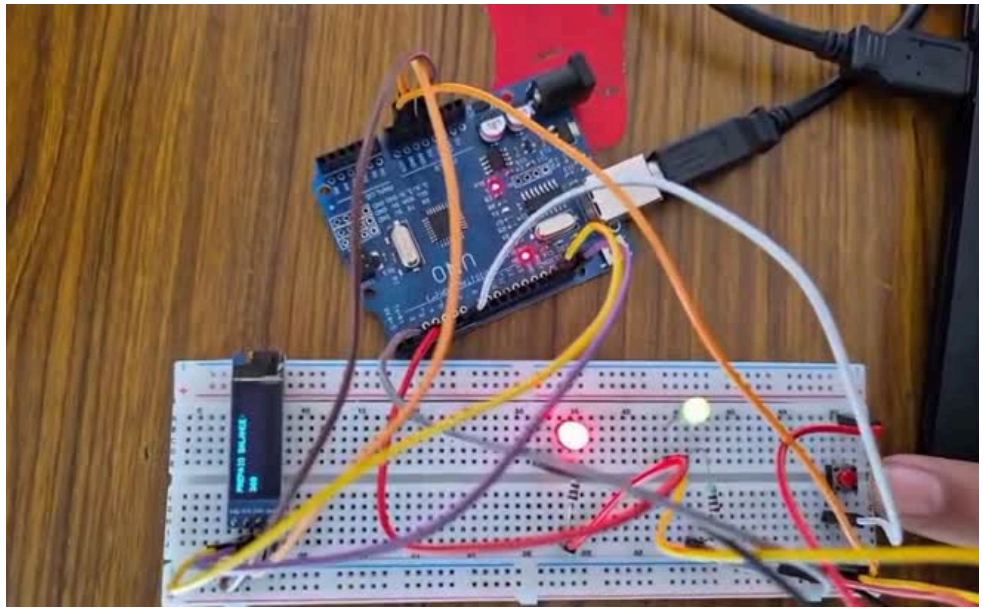
  display.clearDisplay();
  display.setTextSize(1);
  display.setTextColor(SSD1306_WHITE);
  display.setCursor(0,0);
  display.print("PREPAID BALANCE:");
  display.setCursor(0,16);
  display.print(balance);
  display.display();
```



VIDEO OF THE WORKING MODEL
https://drive.google.com/drive/folders/1oZGa9ltSwbFivO3oTOBiAkdkEucUpX3H?usp=sharing







THE INTERVIEW: A SMART METER CONVERSATION

INTERVIEW WITH GENERAL MANAGER OF JHARKHAND ELECTRICITY BOARD

https://drive.google.com/drive/folders/1mHPIE5Y4VJH4SjcK7UDKKiEBYaEE0Sj0?usp=drive_link



SURVEY QUESTIONNAIRE

The Smart Meter Survey

This survey aims to understand consumers' awareness and interest in installing Smart Electricity Meters for homes and small businesses. Your responses will help us design better billing systems, subsidy options, and consumer-friendly policies. The survey takes only a few minutes, and all responses will remain confidential.

What type of electricity connection do you have? — Multiple choice

- House (Domestic)
- Shop
- Small office
- Other small business
- Add option or add "Other"

How much is your average monthly electricity bill? *

- Below ₹1,000
- ₹1,000 – ₹3,000
- ₹3,000 – ₹6,000
- Above ₹6,000
- Not sure

Are you satisfied with your current electricity billing system? *

- Yes, fully satisfied
- Somewhat satisfied
- Not satisfied
- Not sure

Have you heard about Smart Meters before? *

- Yes, I know about them
- I have heard the name only
- No, I have never heard about them

What feature of a Smart Meter would interest you the most? *

- Real-time bill tracking on mobile
- No manual meter reading
- Accurate billing
- Prepaid recharge option
- Not interested

Would you be willing to install a Smart Meter if it helps reduce billing errors? *

- Yes
- Maybe
- No

Would you prefer: *

- Prepaid electricity (like mobile recharge)
- Postpaid monthly bill (current system)
- Both options should be available

What would encourage you to adopt Smart Meter? *

- Discount on electricity bill
- Government subsidy
- Easy mobile app access
- Assurance of no extra hidden charges
- Compulsory government rule

If Smart Meter installation costs ₹1,000–₹2,000, would you: *

- Pay full amount
- Prefer EMI/installment option
- Prefer government subsidy
- Not willing to pay

Do you face any of these issues? *

- High bills
- Billing mistakes
- Power cuts
- No major issues

SURVEY RESPONSES

The Smart Meter Survey

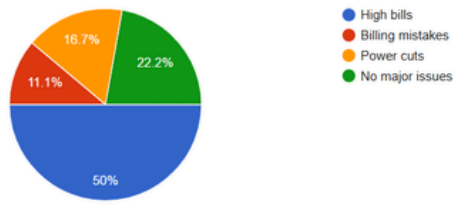
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Sign in to Google to save your progress. Learn more

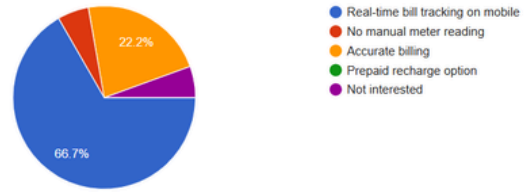
* Indicates required question

SURVEY FORM LINK

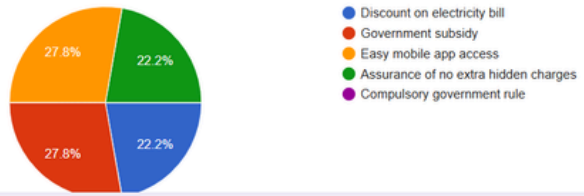
Do you face any of these issues? Copy chart



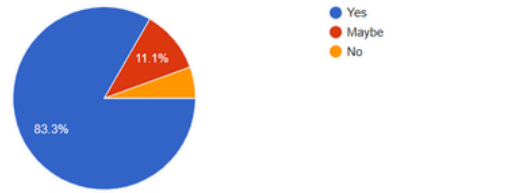
What feature of a Smart Meter would interest you the most? Copy chart



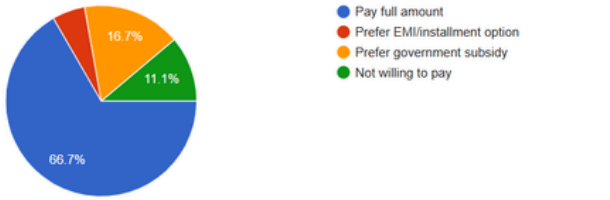
What would encourage you to adopt Smart Meter? Copy chart



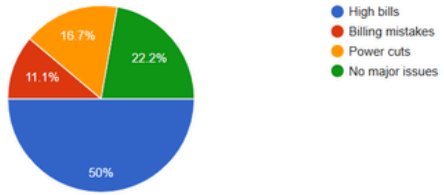
Would you be willing to install a Smart Meter if it helps reduce billing errors? Copy chart



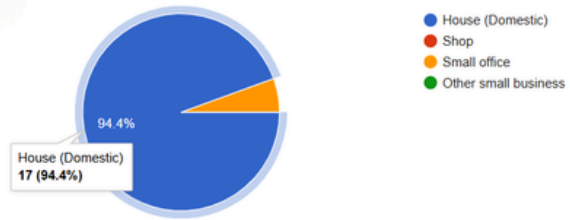
If Smart Meter installation costs ₹1,000–₹2,000, would you: Copy chart



Do you face any of these issues? Copy chart



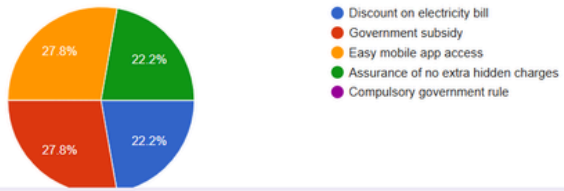
What type of electricity connection do you have? Copy chart



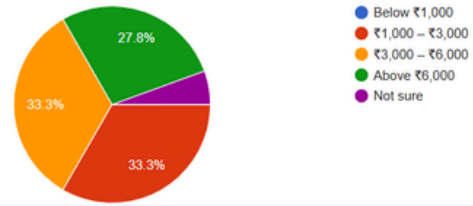
Would you prefer: Copy chart



What would encourage you to adopt Smart Meter? Copy chart



How much is your average monthly electricity bill? Copy chart



TRANSMISSION AND DISTRIBUTION LOSSES OF OTHER COUNTRIES

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Latin America & Caribbean (excluding high income)	2023	14
Least developed countries: UN classification	2022	11
Middle East, North Africa, Afghanistan & Pakistan	2023	5
Middle East, North Africa, Afghanistan & Pakistan (excluding high income)	2022	20
North America	2023	5
OECD members	2023	5
Other small states	2023	6
Pacific island small states		
Small states	2023	6
South Asia	2022	15
Sub-Saharan Africa	2023	24
Sub-Saharan Africa (excluding high income)	2023	24

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World	2023	6
Arab World	2022	16
Caribbean small states	2022	6
Central Europe and the Baltics	2023	6
East Asia & Pacific	2022	4
East Asia & Pacific (excluding high income)	2022	4
Euro area	2023	6
Europe & Central Asia	2023	7
Europe & Central Asia (excluding high income)	2023	9
European Union	2023	6
Fragile and conflict affected situations	2022	28
Heavily indebted poor countries (HIPC)	2022	18
Latin America & Caribbean	2023	13

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Channel Islands		
Chile	2023	6
China	2023	3
Colombia	2023	8
Comoros		
Congo, Dem. Rep.	2022	9
Congo, Rep.	2022	45
Costa Rica	2023	9
Cote d'Ivoire	2022	15
Croatia	2023	10
Cuba	2022	22
Curacao	2022	11
Cyprus	2023	3

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Tanzania	2022	19
Thailand	2023	7
Timor-Leste		
Togo	2022	25
Tonga		
Trinidad and Tobago	2022	5
Tunisia	2022	22
Turkiye	2023	9
Turkmenistan	2022	10
Turks and Caicos Islands		
Tuvalu		
Uganda	2022	19
Ukraine	2023	10

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United Arab Emirates	2022	5
United Kingdom	2023	10
United States	2023	5
Uruguay	2022	9
Uzbekistan	2022	5
Vanuatu		
Venezuela, RB	2022	31
Viet Nam	2022	6
Virgin Islands (U.S.)		
West Bank and Gaza		
Yemen, Rep.	2022	19
Zambia	2022	11
Zimbabwe	2022	22

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Brazil	2023	15
British Virgin Islands		
Brunei Darussalam	2022	9
Bulgaria	2023	6
Burkina Faso	2022	12
Burundi		
Cabo Verde		
Cambodia	2022	14
Cameroon	2022	22
Canada	2023	4
Cayman Islands		
Central African Republic		
Chad	2022	29

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Iraq	2022	59
Ireland	2023	8
Isle of Man		
Israel	2023	4
Italy	2023	7
Jamaica	2022	27
Japan	2023	5
Jordan	2022	11
Kazakhstan	2023	10
Kenya	2023	24
Kiribati		
Korea, Dem. People's Rep.	2022	15
Korea, Rep.	2023	3

Source- https://data.worldbank.org/indicator/EG.ELC.LOSS.ZS?locations=CN%22

Micronesia, Fed. Sts.		
Moldova	2023	9
Monaco		
Mongolia	2022	14
Montenegro	2023	11
Morocco	2023	18
Mozambique	2022	17
Myanmar	2022	8
Namibia	2022	39
Nauru		
Nepal	2022	15

FINANCIALS AND REPORTS

Sr No	States/ Discom	Total Consumer (Nos)	Total Sales (MU)	Domestic Consumer (Nos)	Domestic Sales (MU)	Domestic Consumer to Total Consumer (in %)	Domestic Sales to Total Sales (in %)
1	Uttar Pradesh	34590860	113477.5	30264377	51847.89	87.49%	45.69%
2	Chhattisgarh	6308418	31649.5	5038845	6,531.72	79.87%	20.64%
3	Bihar	20197076	32306.17	17665730	16667.85	87.47%	51.59%
4	JBVNL	5028088	9994.7	4631102	5523.78	92.10%	55.27%
5	Tata Steel Ltd	51221	2976.46	39688	197.04	77.48%	6.62%
6	SAIL, Bokaro	34143	1618.7	31497	101.88	92.25%	6.29%
7	TSUIL	8792	1088.4	6797	30.26	77.31%	2.78%
8	DVC	215	9765.76	5	34.19	2.33%	0.35%

Source: As Per approved tariff order from state Commissions for true up for FY 23-24



	Smart metering	Loss Reduction	Total
Andaman and Nicobar Islands	54	462	516
Andhra Pradesh	4,128	10,708	14,836
Arunachal Pradesh	184	1,042	1,226
Assam	4,050	3,395	7,444
Bihar	2,021	10,559	12,581
Chhattisgarh	4,105	4,021	8,126
Delhi	13	324	337
Goa	469	247	716
Gujarat	10,642	6,089	16,731
Haryana	0	6,794	6,794
Himachal Pradesh	1,788	2,327	4,116
Jammu & Kashmir	1,064	5,034	6,098
Jharkhand	858	3,468	4,326
Karnataka	0	45	45
Kerala	8,231	3,108	11,339
Ladakh	0	876	876
Madhya Pradesh	8,911	9,738	18,649
Maharashtra	15,215	17,238	32,453
Manipur	121	627	748
Meghalaya	310	1,232	1,542
Mizoram	182	322	503
Nagaland	208	466	674
Puducherry	251	84	335
Punjab	5,769	3,873	9,642
	9,715	18,693	28,408
	97	420	518
	19,235	9,568	28,803
	0	120	120
	319	598	917
	18,956	21,960	[illegible]
	1,106	6,100	[illegible]
	12,670	5,000	[illegible]

Table 2. Data of transmission line for 1st cluster

1st cluster				
From	To	Distance (KM)	R (P.U.)	X (P.U.)
1	2	42.8	0.0007704	0.00823044
2	3	18	0.000324	0.0034614
3	4	29.9	0.0005382	0.00574977
4	5	30.9	0.0005562	0.00594207
1	6	27.6	0.0004968	0.00530748
6	7	18	0.000324	0.0034614
7	8	28.4	0.0005112	0.00546132
6	9	21	0.000378	0.0040383
9	10	4.96	0.00008928	0.00095381
10	11	24.7	0.0004446	0.00474981
11	12	28.2	0.0005076	0.00542286
12	13	14.3	0.0002574	0.00274989
13	14	19.5	0.000351	0.00374985
1	15	55.4	0.0009972	0.01065342
15	16	13.2	0.0002376	0.00253836

Table 4. Voltage magnitude and angle for 1st cluster

1st cluster						
Bus	Type	Vsp	PGi	QGi	PLi	QLi
1	2	1.0	59.09	22.18	0	0
2	3	1.0	0	0	7.13	4.06
3	3	1.0	0	0	6.1	3.06
4	3	1.0	0	0	8.09	6.04
5	3	1.0	0	0	7.12	5.08
6	3	1.0	0	0	7.06	4.03
7	3	1.0	0	0	6.06	4.02
8	3	1.0	0	0	5.20	3.10
9	3	1.0	0	0	7.20	5.10
10	3	1.0	0	0	9.06	5.02
11	3	1.0	0	0	7.06	4.02
12	3	1.0	0	0	3.045	1.03
13	3	1.0	0	0	6.06	3.035
14	3	1.0	0	0	5.06	2.035
15	3	1.0	0	0	8.12	7.08
16	3	1.0	0	0	6.12	4.6

Progress Status under RDSS: Physical Progress

Name of Scheme/Project: Revamped Distribution Sector Scheme (RDSS)				
Sl. No.	Particulars	Unit	Awarded Quantity	Executed Quantity
A	**Cabling related works (Replacement)**	cKm	26516	17554
B	**HVDS**			
1	11 KV Line with Dog/Rabbit conductor	cKm	5287	660
2	New 63 KVA DTR	Nos	2730	1640
	Cabling related works (New)			
1	AB Cable of Size 3C*50+1*35 Sq.mm	cKm	2617	1031
C	**Feeder Bifurcation**			
1	11 KV Line with Dog /rabbit conductor	cKm	2200	1040
2	New 63/100/200/500 KVA DTR	Nos	2297	1078
3	Cabling related works (New)	cKm	3320	935
D	**Segregation of Agriculture/Mix Feeders**			
1	11 KV Line with Dog/Rabbit conductor	cKm	1066	131
2	New 100/63 KVA DTR	Nos	448	39
3	AB Cable of Size 3C*50+1*35 Sq.mm	CKm	502	46
E	**Any other Item**			
1	11 KV VCB	SET	558	68
2	11 KV AB Switch	SET	4478	631

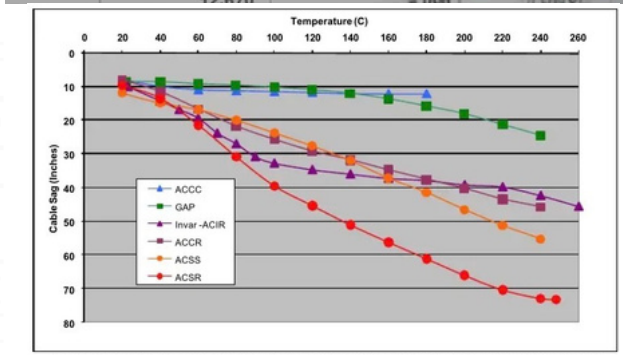


Figure 6 - Sag and Temperature Comparison between Different Types of Conductors (1600 amps)

SOURCES AND RESEARCH PAPERS

TRANSMISSION LOSSES REPORT

- https://www.canva.com/design/DAHCr6PpIPY/EDvaYZ-tvmvP4FqQw5PaNg/view
- https://www.canva.com/design/DAHCr25Xj-Y/SkSiHwyOaI9djloZEFxADg/view

GOOGLE BASED ANTI THEFT REPORT

- https://www.canva.com/design/DAHCr7KJEvs/rW7jA3wRI1AkQfyChhz8Tg/view
- https://www.canva.com/design/DAHCr3Aiamk/KbuqFzJQGhKMxIQMFXSFfg/view

MARKET ANALYSIS REPORT

- https://docs.google.com/document/d/12QxftQxyblC9XbY7ufnEAw4UEKViv G-8mjT9m4sIb2Q/edit?usp=sharing

THANK YOU

Drive Link to the Video pitch, Interview and the working model:

https://drive.google.com/drive/folders/1vJhKGfcJCMaTxCBy9zdG5nWJPl2TTAMM

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